

Speed Ramps Testing

Start from a certain speed, command positive acceleration and wait until speed is reached.
Hold at steady state for 10 seconds then command deceleration.

Speed [mph]	Acceleration [mps ²]	Completion Check	Date	Notes	Test #
0-20 mph	±0.25	x	9/16	ok	62409002
	±0.5	x	9/16	ok	62409002
	±0.75	x	9/16	ok	62409002
	±1	x	9/16	ok	62409002
	±1.5	x	9/16	ok	62409002
	±2	x	9/16	Ok, this is the max/min acceleration at this speed	62409002
	±2.25			System fault	
	±3				

Speed [mph]	Acceleration [mps ²]	Completion Check	Date	Notes	Test #
10-30 mph	±0.25	x	9/16/24	ok	624009009
	±0.5	x	9/16/24	ok	624009009
	±0.75	x	9/16/24	ok	624009009
	±1	x	9/16/24	ok	624009009
	±1.5	x	9/16/24	ok	624009009
	±2	x	9/16/24	ok	624009009
	±2.01 (1 st)	x	9/16/24	Vehicle came down to a stop by itself	624009009
	±2.01 (2 nd)	x	9/16/24	System faulted	624009009

Speed [mph]	Acceleration [mps ²]	Completion Check	Date	Notes	Test #
20-40 mph	±0.25	x	9/16	- Vehicle takes some time to start from 0, need to look into data to know how much - To maintain speed @ 40mph you need to send some acceleration (0.03 mps²)	624009003
	±0.5	x	9/16	ok	624009003
	±0.75	x	9/16	ok	624009003
	±1	x	9/16	ok	624009003
	±1.5	x	9/16	ok	624009003
	±2	x	9/16	Ok, this is the max/min acceleration at this speed	624009003
	±2.25			System fault	
	±3				

Speed [mph]	Acceleration [mps ²]	Completion Check	Date	Notes	Test #
30-50 mph	±0.25	x	9/16/2024	To maintain 30 mph send 0.025 mps ² To maintain 50 mph send 0.08 mps ²	624009010
	±0.5	X	9/16/2024	ok	624009010
	±0.75	X	9/16/2024	ok	624009010
	±1	x	9/16/2024	ok	624009010
	±1.5	x	9/16/2024	ok	624009010
	±2	x	9/16/2024	ok	624009010
	±2.01	x	9/16/2024	System fault	624009010
	±3				

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
40-60 mph	±0.25	x	9/16/2024	To maintain 60 mph, send 0.125 mps2 To maintain 40 mph, send 0.035 mps2	624009004
	±0.5	x	9/16/2024	ok	624009004
	±0.75	x	9/16/2024	ok	624009004
	±1	x	9/16/2024	ok	624009004
	±1.5	x	9/16/2024	ok	624009004
	±2	x	9/16/2024	Ok, this is the max/min acceleration at this speed	624009004
	±2.25			System fault	
	±3				

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
50-70 mph	±0.25	x	9/16/2024	To maintain 50 mph send 0.08 mps2 To maintain 70 mph send 0.19 mps2	624009005
	±0.5	x	9/16/2024	Done twice	624009005
	±0.75	x	9/16/2024	ok	624009005
	±1	x	9/16/2024	ok	624009005
	±1.5	x	9/16/2024	ok	624009005
	±2	x	9/16/2024	Ok, this is the max/min acceleration at this speed	624009005
	±2.1			System fault	624009005
	±3				624009005

Acceleration Limit Testing

Start from a certain speed, command acceleration, check system functionalities

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
Start @0mph	±1	x	9/16/2024	ok	624009007
	±1.5	x	9/16/2024	ok	624009007
	±1.8	x	9/16/2024	Done twice	624009007
	±1.83	x	9/16/2024	Ok, done twice	624009007
	+1.84	x	9/16/2024	Went up to ~55 mph then clipped off and started decelerating	624009007
	+1.85	x	9/16/2024	Went up to ~60 mph then clipped off and started decelerating	624009007
	+1.89	x	9/16/2024	Went up to ~55 mph then clipped off and started decelerating	624009007
	+1.9	x	9/16/2024	Went up to ~50 mph then clipped off and started decelerating	624009007
	+2	x	9/16/2024	Went up to ~50 mph then clipped off and started decelerating	624009007
	±2.1				

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
Start @60mph	±1	x	9/16/2024	Done twice	624009006
	±1.5	x	9/16/2024	ok	624009006
	±2	x	9/16/2024	Ok, this is the max/min acceleration at this speed	624009006
	±2.1			System fault	

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
Start @65mph	±1				
	±1.5				
	±2	x	9/16/2024	Ok, this is the max/min acceleration at this speed	624009006
	±2.1			System fault	

Deceleration Limit Testing

Start from a certain speed, command deceleration, check system functionalities

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
Start @20mph	-2	x	9/16/2024	ok	624009011
	-2.01	x	9/16/2024	ok	624009011
	-2.1	x	9/16/2024	ok	624009011
	-2.5	x	9/16/2024	ok	624009011

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
Start @40mph	-2	x	9/16/2024	ok	624009012
	-2.5	x	9/16/2024	ok	624009012
	-3	x	9/16/2024	ok	624009012

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
Start @60mph	-2	x	9/16/2024	ok	624009013
	-2.5	x	9/16/2024	ok	624009013
	-3	x	9/16/2024	Ok, but tire pressure sensor and parking brake faulted	624009013
	-3.5				

Starting from Stop Testing

Speed [mph]	Acceleration [mps2]	Completion Check	Date	Notes	Test #
0mph	±0.5 (no resume)	x	9/16/24	Time: ~3sec	624009008
	±0.5 (with resume pedal)	x	9/16/24	Kicks out of openpilot override when pedal is pressed	624009008
	±0.5 (resume button)	x	9/16/24	Done twice, time: ~2 sec	624009008
	±0.5 (no resume)- 2 nd trial	x	9/16/24	Time : ~2.40 sec	624009008
	±0.5 (resume button) 2 nd trial	x	9/16/24	Time : ~2 sec	624009008